

Every table, and what it is for

What each table holds, every column it has, and how it joins to the rest. Columns, types, keys and row counts are read from the running database rather than written by hand, so they cannot drift from what Postgres actually enforces.

86	990	226	58	28
tables	columns	foreign keys	holding data	still empty

Generated against migration `c3f61e0a77d2` on branch `slice/office-feedback`. Row counts come from the seeded demo school and will differ on another installation.

How to read it

Five rules explain why the tables look the way they do. Without them the list that follows is just nouns. They are numbered because the rest of the documentation refers to them by number.

1 Almost every table carries `school_id`

The product is sold to separate, independent schools; a tenant is a customer, not a branch. The column is declared on `TenantBase` so a new table cannot quietly be created without one. Three tables derive from `TimestampedBase` instead and have none: `schools` is the tenant, and `permissions` and `scheduled_jobs` belong to the software rather than to any one school.

2 Year-scoped facts hang off `enrolment_id`

A fee, an attendance mark, a bus seat and a set of marks belong to a child's year in a class, so they point at `enrolments`. A name and a date of birth belong to the child for life, so they point at `students`.

3 Money is never edited

An invoice is voided and reissued; a payment is reversed by a contra entry with a negative amount and `reverses_payment_id` pointing at the original.

4 A balance is always a SUM

Payments allocate to invoice lines through `payment_allocations`, oldest first. What a family owes is computed every time it is asked for.

5 Destructive actions carry a reason

`void`, `status_change` and `delete` refuse to commit without one, and what is stored is the words the person typed.

What is in here

AREA	IN USE	EMPTY
Tenancy, identity and access	11	0
People - students and guardians	4	0
People - staff	2	0
Academics	8	1
Attendance	1	1
Examinations and results	7	1
Fees and money	8	2
Transport	5	0
HR and payroll	3	7
Admission	3	12

AREA	IN USE	EMPTY
Communication	2	3
System	4	1

An empty table is not an unfinished one. Each belongs to a feature that is built and has working endpoints; this demo school simply has not used it.

Part one: tables in use

Every table here holds data in the seeded school, and the row counts are live.

Tenancy, identity and access

schools 2 rows

One customer. The only table without a `school\_id` of its own.

Pointed at by academic_years, admission_cycles, admission_decisions, admission_offers, application_guardians, application_medical, application_payments, application_siblings, applications, assessment_schemes, assessment_subjects, assessments, attendance, audit_log, class_sections, class_subject_teacher, custom_fields, cycle_class_config, departments, document_types, documents, employees, enquiries, enquiry_interactions, enrolments, exam_schedule, exams, fee_concessions, fee_heads, fee_invoice_lines, fee_invoices, fee_payments, fee_periods, fee_plan_items, fee_plans, grade_bands, grading_scales, guardians, holidays, homework, homework_submissions, interviews, jobs, leave_balances, leave_types, marks, message_recipients, message_templates, messages, notices, notification_preferences, number_sequences, payment_allocations, payroll_runs, payslip_lines, payslips, report_card_publications, role_permissions, roles, route_stops, routes, salary_components, salary_structure_items, salary_structures, scheme_components, school_periods, settings, staff_attendance, staff_leave_requests, student_fee_plans, student_guardian, student_leave_requests, students, subjects, substitutions, timetable_slots, transport_assignments, transport_fee_slabs, user_roles, users, vehicles, waitlist_entries

Unique on code

COLUMN	TYPE	NULL	NOTES
code	varchar(16)		unique
name	varchar(160)		
status	varchar(10)		default 'active'
address	text	nullable	
city	varchar(80)	nullable	
state	varchar(80)	nullable	
pincode	varchar(10)	nullable	
phone	varchar(20)	nullable	
email	varchar(160)	nullable	
website	varchar(200)	nullable	
logo_url	text	nullable	
primary_color	varchar(9)	nullable	
board	varchar(20)	nullable	
affiliation_no	varchar(40)	nullable	
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

academic_years

2 rows

A session. Several may be open at once — 2026-27 active while 2027-28 takes admissions and 2025-26 is closing is the normal state of a school in January, which the v0 string could not express.

Points at schools

Pointed at by admission_cycles, assessment_schemes, audit_log, class_sections, enrolments, fee_invoices, fee_plans, holidays, leave_balances, staff_leave_requests

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
code	varchar(9)		unique
start_date	date		
end_date	date		
status	varchar(15)		default 'planning'
is_current	boolean		default false
promotion_completed_at	date	nullable	
result_published_at	date	nullable	
closed_at	date	nullable	
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

users216 rows

Every person who can sign in - staff, students and guardians alike. One row per login; the role-specific detail lives in `students`, `guardians` or `employees`, each of which points back here.

Points at schools

Pointed at by admission_decisions, application_payments, applications, assessments, audit_log, documents, employees, enquiries, enquiry_interactions, exam_schedule, fee_concessions, fee_payments, fee_periods, guardians, interviews, marks, message_recipients, messages, notices, notification_preferences, payroll_runs, report_card_publications, staff_attendance, staff_leave_requests, student_fee_plans, student_leave_requests, students, user_roles

Unique on school_id,login_id

COLUMN	TYPE	NULL	NOTES
role	varchar(7)		
login_id	varchar(64)		unique
password_hash	varchar(255)		
full_name	varchar(120)		
email	varchar(160)	nullable	
phone	varchar(20)	nullable	
photo_url	text	nullable	
is_active	boolean		
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id, unique

roles28 rows

A named set of permissions. System roles ship with the product and cannot be deleted; a school may copy one and adjust it.

Points at schools

Pointed at by role_permissions, user_roles

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
code	varchar(40)		unique
name	varchar(80)		
description	text	nullable	
is_system	boolean		default false
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

permissions 77 rows

A thing that can be done, named `module.resource.action`.

Pointed at by role_permissions

Unique on code

COLUMN	TYPE	NULL	NOTES
code	varchar(80)		unique
module	varchar(40)		
action	varchar(40)		
description	text	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

role_permissions 293 rows

Which permissions a role grants. Per school, so two tenants can define the same role differently.

Points at permissions, roles, schools

Unique on role_id,permission_id

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
role_id	bigint		→ roles.id, unique
permission_id	bigint		→ permissions.id, unique
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

user_roles 223 rows

A user holds a role, optionally narrowed to part of the school.

Points at roles, schools, users

Unique on user_id,role_id,scope_type,scope_id

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
user_id	bigint		→ users.id, unique
role_id	bigint		→ roles.id, unique
scope_type	varchar(13)		unique, default 'school'
scope_id	bigint	nullable	unique
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

settings2 rows

Per-school configuration as key/value pairs, with the vocabulary defined in `core/settings\_registry.py`. Module switches live here too, as `feature`.

Points at schools
Unique on school_id,key

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
key	varchar(80)		unique
value	json		
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

custom_fields2 rows

One school-defined attribute on students, guardians, employees or applications (§3.15 level 2).

Points at schools
Unique on school_id,entity,key

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
entity	varchar(11)		unique
key	varchar(40)		unique
label	varchar(120)		
field_type	varchar(7)		
options	json	nullable	
is_required	boolean		default false
is_active	boolean		default true
sort_order	integer		default 100
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

number_sequences

3 rows

A gapless counter per school, per type, per year.

Points at schools

Unique on school_id,kind,year

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
kind	varchar(24)		unique
year	integer		unique
prefix	varchar(24)	nullable	
width	integer		default 6
next_value	integer		default 1
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

alembic_version

1 rows

Alembic's own bookkeeping - the single row naming the migration this database is at. Not part of the product; never edit it by hand.

COLUMN	TYPE	NULL	NOTES
version_num	varchar(32)		

People – students and guardians

students

100 rows

What is true about a student for life. Which class they sit in is a fact about a *year* and lives on `enrolments` (ERP_BLUEPRINT §3.2).

Points at schools, users

Pointed at by application_siblings, applications, enrolments, homework_submissions, marks, message_recipients, student_guardian

Unique on user_id · school_id, admission_no

COLUMN	TYPE	NULL	NOTES
user_id	bigint		→ users.id, unique
admission_no	varchar(32)		unique
dob	date	nullable	
gender	varchar(6)	nullable	
address	text	nullable	
admission_date	date	nullable	
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id, unique
status	varchar(15)		default 'active'
custom	json		default '{}'

guardians

98 rows

Whoever is responsible for a child — not necessarily a parent (§3.4).

Points at schools, users

Pointed at by message_recipients, student_guardian

Unique on user_id

COLUMN	TYPE	NULL	NOTES
user_id	bigint		→ users.id, unique
occupation	varchar(80)	nullable	
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id
employee_id	bigint	nullable	

student_guardian 100 rows

The link, with the structure §3.4 asks for: which relation, and which one of them the school actually rings.

Points at guardians, schools, students

Unique on guardian_id,student_id

COLUMN	TYPE	NULL	NOTES
guardian_id	bigint		→ guardians.id, unique
student_id	bigint		→ students.id, unique
relation	varchar(14)		
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id
is_primary	boolean		default false

enrolments 100 rows

A student's membership of a section for one academic year. ****The most important join in the schema****: every year-scoped fact - fees, attendance, marks, a bus seat - hangs off `enrolment\_id`, never `student\_id`.

Points at academic_years, class_sections, schools, students

Pointed at by attendance, fee_concessions, fee_invoices, fee_payments, report_card_publications, student_fee_plans, student_leave_requests, transport_assignments

Unique on class_section_id,roll_no

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
student_id	bigint		→ students.id
academic_year_id	bigint		→ academic_years.id
class_section_id	bigint		→ class_sections.id, unique
roll_no	integer		unique
status	varchar(15)		default 'active'
joined_on	date	nullable	
left_on	date	nullable	
house	varchar(20)	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

People - staff

employees 16 rows

Any member of staff, teaching or not (ERP_BLUEPRINT §3.4).

Points at departments, employees, schools, users

Pointed at by application_guardians, attendance, class_sections, class_subject_teacher, departments, homework, leave_balances, payslips, routes, salary_structures, staff_attendance, staff_leave_requests, substitutions, timetable_slots

Unique on user_id · school_id,employee_code

COLUMN	TYPE	NULL	NOTES
user_id	bigint		→ users.id, unique
employee_code	varchar(16)		unique
qualification	varchar(120)	nullable	
joining_date	date	nullable	
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id, unique
employee_type	varchar(14)		default 'teaching'
department_id	bigint	nullable	→ departments.id
reporting_to_id	bigint	nullable	→ employees.id
status	varchar(13)		default 'active'
exited_on	date	nullable	
designation	varchar(60)	nullable	
emergency_contact_name	varchar(120)	nullable	
emergency_contact_phone	varchar(20)	nullable	
pan	varchar(10)	nullable	
uan	varchar(12)	nullable	
esi_number	varchar(20)	nullable	
bank_account_no	varchar(20)	nullable	
bank_ifsc	varchar(11)	nullable	
bank_name	varchar(80)	nullable	

departments

6 rows

A teaching or administrative department (§5.3.3).

Points at employees, schools

Pointed at by employees

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
code	varchar(12)		unique
name	varchar(80)		
head_employee_id	bigint	nullable	→ employees.id
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Academics

class_sections 10 rows

One class-and-section for one academic year - 10-A, 9-B - with its capacity, room and class teacher.
The unit almost everything academic is scoped to.

Points at academic_years, employees, schools

Pointed at by class_subject_teacher, enrolments, exam_schedule, homework, notices, timetable_slots

Unique on academic_year_id,class_name,section

COLUMN	TYPE	NULL	NOTES
class_name	varchar(8)		unique
section	varchar(4)		unique
class_teacher_id	bigint	nullable	→ employees.id
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id
academic_year_id	bigint		→ academic_years.id, unique
capacity	integer	nullable	
stream	varchar(20)	nullable	
room	varchar(20)	nullable	

subjects 9 rows

The catalogue of subjects the school teaches. A subject exists once and is attached to sections through
`class\_subject\_teacher`.

Points at schools

Pointed at by class_subject_teacher, exam_schedule, homework, timetable_slots

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
name	varchar(60)		
code	varchar(12)		unique
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id, unique

class_subject_teacher

60 rows

Which teacher teaches which subject to which section. The join that makes a timetable solvable and decides whose marks entry is allowed.

Points at class_sections, employees, schools, subjects

Unique on class_section_id,subject_id

COLUMN	TYPE	NULL	NOTES
class_section_id	bigint		→ class_sections.id, unique
subject_id	bigint		→ subjects.id, unique
teacher_id	bigint		→ employees.id
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id

school_periods

6 rows

Bell timings, once per school instead of once per slot.

Points at schools

Pointed at by timetable_slots

Unique on school_id,period_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
period_no	integer		unique
start_time	time without time zone		
end_time	time without time zone		
name	varchar(20)	nullable	
is_break	boolean		default false
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

timetable_slots

300 rows

The weekly grid. Which subject, taught by which teacher, occupies which period on which weekday for which section.

Points at class_sections, employees, school_periods, schools, subjects

Pointed at by substitutions

Unique on class_section_id,day_of_week,period_id

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
class_section_id	bigint		→ class_sections.id, unique
day_of_week	varchar(3)		unique
period_id	bigint		→ school_periods.id, unique
subject_id	bigint		→ subjects.id
teacher_id	bigint		→ employees.id
room	varchar(20)	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

holidays

5 rows

A day the school is shut.

Points at academic_years, schools

Unique on academic_year_id,date

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
academic_year_id	bigint		→ academic_years.id, unique
date	date		unique
name	varchar(80)		
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

homework

8 rows

A piece of homework set for a section and subject, with the date it is due.

Points at class_sections, employees, schools, subjects

Pointed at by homework_submissions

COLUMN	TYPE	NULL	NOTES
class_section_id	bigint		→ class_sections.id
subject_id	bigint		→ subjects.id
teacher_id	bigint		→ employees.id
title	varchar(160)		
description	text	nullable	
assigned_date	date		
due_date	date		
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id

homework_submissions

47 rows

A child's submission against one `homework` row, and whether it has been marked.

Points at homework, schools, students

Unique on homework_id,student_id

COLUMN	TYPE	NULL	NOTES
homework_id	bigint		→ homework.id, unique
student_id	bigint		→ students.id, unique
answer_text	text		
submitted_at	timestampz		
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id

Attendance

attendance 5,800 rows

One day, one child, one mark.

Points at employees, enrolments, schools

Unique on enrolment_id,date

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id, unique
date	date		unique
status	varchar(8)		
marked_by	bigint	nullable	→ employees.id
remarks	varchar(200)	nullable	
corrected_by	bigint	nullable	→ employees.id
corrected_at	timestamptz	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Examinations and results

exams 4 rows

One assessment event: "Term 1 Periodic Test", "Term 1 Examination".

Points at scheme_components, schools

Pointed at by exam_schedule

COLUMN	TYPE	NULL	NOTES
name	varchar(120)		
term	varchar(20)		
start_date	date		
end_date	date		
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id
scheme_component_id	bigint	nullable	→ scheme_components.id

exam_schedule 240 rows

One paper: this exam, this section, this subject.

Points at class_sections, exams, schools, subjects, users

Pointed at by marks

Unique on exam_id,class_section_id,subject_id

COLUMN	TYPE	NULL	NOTES
exam_id	bigint		→ exams.id, unique
class_section_id	bigint		→ class_sections.id, unique
subject_id	bigint		→ subjects.id, unique
exam_date	date		
start_time	time without time zone	nullable	
max_marks	numeric(5,2)		
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id
room	varchar(20)	nullable	
marks_locked_at	timestamptz	nullable	
marks_locked_by	bigint	nullable	→ users.id

marks 2,400 rows

One child's result on one paper.

Points at exam_schedule, schools, students, users

Unique on exam_schedule_id,student_id

COLUMN	TYPE	NULL	NOTES
exam_schedule_id	bigint		→ exam_schedule.id, unique
student_id	bigint		→ students.id, unique
marks_obtained	numeric(5,2)	nullable	
entered_by	bigint		→ users.id
id	bigint		primary key
created_at	timestamptz		default CURRENT_TIMESTAMP
updated_at	timestamptz		default now()
school_id	bigint		→ schools.id
is_absent	boolean		default false
is_exempted	boolean		default false
remarks	varchar(200)	nullable	

assessment_schemes 1 rows

The shape of a year's assessment: its terms and what each is marked out of (§5.4.3).

Points at academic_years, schools

Pointed at by report_card_publications, scheme_components

Unique on academic_year_id,name

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
academic_year_id	bigint		→ academic_years.id, unique
name	varchar(80)		unique
is_active	boolean		default false
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

scheme_components

8 rows

One markable column of the report card: "Term 1, Periodic Test, out of 10".

Points at assessment_schemes, schools

Pointed at by exams

Unique on scheme_id,term,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
scheme_id	bigint		→ assessment_schemes.id, unique
term	varchar(20)		unique
code	varchar(12)		unique
name	varchar(60)		
max_marks	numeric(5,2)		
sequence	integer		default 0
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

grading_scales

1 rows

A named set of grade bands, versioned.

Points at schools

Pointed at by grade_bands, report_card_publications

Unique on school_id,name,version

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
name	varchar(60)		unique
version	integer		unique, default 1
is_active	boolean		default false
frozen_at	timestamptz	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

grade_bands

8 rows

One row of a grading scale: "91 and above is an A1".

Points at `grading_scales`, `schools`

Unique on `grading_scale_id,min_percent` · `grading_scale_id,grade`

COLUMN	TYPE	NULL	NOTES
min_percent	numeric(5,2)		unique
grade	varchar(4)		unique
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id
grading_scale_id	bigint		→ grading_scales.id, unique
description	varchar(40)	nullable	

Fees and money

fee_heads

5 rows

The named charges a school bills: tuition, transport, exam fee. The vocabulary every invoice line is built from.

Points at schools

Pointed at by fee_concessions, fee_invoice_lines, fee_plan_items

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
name	varchar(60)		
code	varchar(16)		unique
type	varchar(9)		
is_refundable	boolean		default false
gl_code	varchar(20)	nullable	
is_active	boolean		default true
created_at	timestampz		default now()
updated_at	timestampz		default now()

fee_plans

10 rows

What a class is charged for a year.

Points at academic_years, schools

Pointed at by fee_plan_items, student_fee_plans

Unique on academic_year_id,name

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
academic_year_id	bigint		→ academic_years.id, unique
name	varchar(60)		unique
class_name	varchar(8)	nullable	
is_active	boolean		default true
created_at	timestampz		default now()
updated_at	timestampz		default now()

fee_plan_items 30 rows

The lines of a fee plan - which head, how much, how often. A plan means nothing without these.

Points at fee_heads, fee_plans, schools

Unique on fee_plan_id,fee_head_id

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
fee_plan_id	bigint		→ fee_plans.id, unique
fee_head_id	bigint		→ fee_heads.id, unique
amount	numeric(10,2)		
frequency	varchar(8)		
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

fee_concessions 2 rows

A discount that someone approved.

Points at enrolments, fee_heads, schools, users

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id
type	varchar(11)		
fee_head_id	bigint	nullable	→ fee_heads.id
percent	numeric(5,2)	nullable	
amount	numeric(10,2)	nullable	
reason	text		
status	varchar(9)		
requested_by	bigint	nullable	→ users.id
approved_by	bigint	nullable	→ users.id
decided_at	timestamptz	nullable	
valid_from	date	nullable	
valid_to	date	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

fee_invoices

300 rows

One month's bill for one enrolment (\$0.6: monthly, per student).

Points at academic_years, enrolments, schools

Pointed at by fee_invoice_lines

Unique on school_id,invoice_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
enrolment_id	bigint		→ enrolments.id
academic_year_id	bigint		→ academic_years.id
invoice_no	varchar(24)		unique
period_month	integer		
period_year	integer		
issued_on	date		
due_date	date		
status	varchar(14)		
settled_on	date	nullable	
void_reason	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

fee_invoice_lines

900 rows

One head on one invoice. The unit a payment allocates against.

Points at fee_heads, fee_invoices, schools

Pointed at by payment_allocations

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
invoice_id	bigint		→ fee_invoices.id
fee_head_id	bigint		→ fee_heads.id
description	varchar(80)		
amount	numeric(10,2)		
discount	numeric(10,2)		default '0'
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

fee_payments

170 rows

Money received. Against an enrolment, not against an invoice.

Points at enrolments, fee_payments, schools, users

Pointed at by payment_allocations

Unique on school_id,idempotency_key · school_id,receipt_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
enrolment_id	bigint		→ enrolments.id
receipt_no	varchar(24)		unique
amount	numeric(10,2)		
method	varchar(20)		
instrument_ref	varchar(40)	nullable	
received_at	timestamptz		
received_by	bigint	nullable	→ users.id
idempotency_key	varchar(64)		unique
status	varchar(8)		
reverses_payment_id	bigint	nullable	→ fee_payments.id
reason	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

payment_allocations

450 rows

How much of a payment settled which line. The ledger's only truth about what is paid — every balance in the system is a SUM over this table.

Points at fee_invoice_lines, fee_payments, schools

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
payment_id	bigint		→ fee_payments.id
invoice_line_id	bigint		→ fee_invoice_lines.id
amount	numeric(10,2)		
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Transport

vehicles 2 rows

A bus. `capacity` is the seating figure on the registration papers, and it is the number the hard block in §5.6.9 counts against — not a soft target the office can talk its way past.

Points at schools

Pointed at by routes

Unique on school_id,registration_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
registration_no	varchar(20)		unique
make_model	varchar(60)	nullable	
capacity	integer		
ownership	varchar(5)		default 'owned'
status	varchar(17)		default 'active'
gps_device_id	varchar(40)	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

routes 2 rows

One bus doing one circuit.

Points at employees, schools, vehicles

Pointed at by route_stops

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
code	varchar(16)		unique
name	varchar(80)		
vehicle_id	bigint	nullable	→ vehicles.id
driver_id	bigint	nullable	→ employees.id
attendant_id	bigint	nullable	→ employees.id
distance_km	numeric(6,2)	nullable	
status	varchar(9)		default 'planned'
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

route_stops

7 rows

A halt on a route, with the times the bus is actually there.

Points at routes, schools, transport_fee_slabs

Pointed at by transport_assignments

Unique on route_id,sequence

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
route_id	bigint		→ routes.id, unique
sequence	integer		unique
name	varchar(80)		
landmark	varchar(120)	nullable	
pickup_time	time without time zone		
drop_time	time without time zone	nullable	
fee_slab_id	bigint	nullable	→ transport_fee_slabs.id
created_at	timestamptz		default now()
updated_at	timestamptz		default now()
latitude	float	nullable	
longitude	float	nullable	

transport_assignments

30 rows

A child on a bus, from a date until a date.

Points at enrolments, route_stops, schools

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id
route_stop_id	bigint		→ route_stops.id
direction	varchar(6)		default 'both'
start_date	date		
end_date	date	nullable	
status	varchar(9)		default 'requested'
end_reason	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

transport_fee_slabs

3 rows

A distance band and its monthly price.

- Points at schools
- Pointed at by route_stops
- Unique on school_id,name

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
name	varchar(40)		unique
monthly_amount	numeric(10,2)		
is_active	boolean		default true
created_at	timestampz		default now()
updated_at	timestampz		default now()

HR and payroll

leave_types 4 rows

One kind of staff leave, with its yearly entitlement.

Points at schools

Pointed at by leave_balances, staff_leave_requests

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
code	varchar(12)		unique
name	varchar(60)		
annual_quota	numeric(5,1)		
is_paid	boolean		default true
active	boolean		default true
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

salary_components 11 rows

One line a payslip can carry, and how its amount is worked out.

Points at schools

Pointed at by payslip_lines, salary_structure_items

Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
code	varchar(16)		unique
name	varchar(60)		
type	varchar(21)		
calculation	varchar(16)		
value	numeric(12,2)		default '0'
applies_below_gross	numeric(12,2)	nullable	
taxable	boolean		default true
statutory	boolean		default false
active	boolean		default true
sequence	integer		default 0
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

salary_structures

16 rows

What one employee is on, from a date.

Points at employees, schools

Pointed at by salary_structure_items

Unique on employee_id,effective_from

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
employee_id	bigint		→ employees.id, unique
effective_from	date		unique
monthly_gross	numeric(12,2)		
is_active	boolean		default false
note	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Admission

enquiries 6 rows

A prospective parent's enquiry, before any application exists. The top of the admission funnel.

Points at admission_cycles, applications, schools, users

Pointed at by enquiry_interactions

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
cycle_id	bigint		→ admission_cycles.id
enquirer_name	varchar(120)		
mobile	varchar(20)		
email	varchar(160)	nullable	
child_name	varchar(120)	nullable	
child_dob	date	nullable	
class_of_interest	varchar(8)	nullable	
source	varchar(10)		default 'walk_in'
status	varchar(23)		default 'new'
assigned_to	bigint	nullable	→ users.id
next_follow_up_on	date	nullable	
converted_application_id	bigint	nullable	→ applications.id
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

admission_cycles

1 rows

One intake season for one academic year.

Points at academic_years, schools

Pointed at by applications, cycle_class_config, enquiries, waitlist_entries

Unique on school_id,academic_year_id,name

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
academic_year_id	bigint		→ academic_years.id, unique
name	varchar(80)		unique
status	varchar(8)		default 'planning'
starts_on	date	nullable	
ends_on	date	nullable	
application_fee	numeric(10,2)		default '0'
late_fee	numeric(10,2)		default '0'
allow_online_applications	boolean		default true
admission_fee_refund_policy	text	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

cycle_class_config

3 rows

Seats and rules for one class within one cycle.

Points at admission_cycles, schools

Unique on cycle_id,class_name,stream

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
cycle_id	bigint		→ admission_cycles.id, unique
class_name	varchar(8)		unique
stream	varchar(20)	nullable	unique
total_seats	integer		default 0
reserved_seats	json	nullable	
age_on	date	nullable	
min_age_years	numeric(4,2)	nullable	
max_age_years	numeric(4,2)	nullable	
requires_test	boolean		default false
requires_interview	boolean		default false
required_document_codes	json	nullable	
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

Communication

notices

6 rows

The noticeboard. An announcement with an audience - the whole school, one class, parents, teachers - optionally also sent as a message.

Points at class_sections, messages, schools, users

COLUMN	TYPE	NULL	NOTES
title	varchar(160)		
body	text		
audience	varchar(8)		
class_section_id	bigint	nullable	→ class_sections.id
published_by	bigint		→ users.id
published_at	timestampz		
id	bigint		primary key
created_at	timestampz		default CURRENT_TIMESTAMP
updated_at	timestampz		default now()
school_id	bigint		→ schools.id
message_id	bigint	nullable	→ messages.id

message_templates

4 rows

Versioned, so the exact text sent stays reproducible (§5.9.9).

Points at schools

Pointed at by messages

Unique on school_id,code,version

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
code	varchar(40)		unique
version	integer		unique, default 1
name	varchar(80)		
channel	varchar(8)		default 'email'
category	varchar(11)		
subject	varchar(200)		
body	text		
is_active	boolean		default true
created_at	timestampz		default now()
updated_at	timestampz		default now()

System

audit_log

264 rows

Append-only. No update path, no delete path, never truncated.

Points at academic_years, schools, users

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
occurred_at	timestamptz		
actor_user_id	bigint	nullable	→ users.id
actor_label	varchar(120)	nullable	
entity_type	varchar(40)		
entity_id	bigint	nullable	
action	varchar(13)		
before	json	nullable	
after	json	nullable	
reason	text	nullable	
academic_year_id	bigint	nullable	→ academic_years.id
ip	varchar(45)	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

documents13 rows

An uploaded file attached to a student, an employee or a vehicle, with its verification state and expiry.
Deleted softly, so the audit entry still points at something.

Points at document_types, schools, users

Unique on file_key

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
owner_type	varchar(11)		
owner_id	bigint		
document_type_id	bigint	nullable	→ document_types.id
file_key	varchar(255)		unique
file_name	varchar(255)		
mime_type	varchar(120)		
size_bytes	bigint		
checksum	varchar(64)	nullable	
uploaded_by	bigint	nullable	→ users.id
uploaded_at	timestamptz		
status	varchar(17)		default 'pending'
original_seen	boolean		default false
verified_by	bigint	nullable	→ users.id
verified_at	timestamptz	nullable	
rejection_reason	text	nullable	
expires_on	date	nullable	
is_confidential	boolean		default false
deleted_at	timestamptz	nullable	
deleted_by	bigint	nullable	
delete_reason	text	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

document_types

42 rows

Configurable, not an enum: required-document lists change by class, by category and by state regulation, and a school must be able to edit them without a deploy (§3.8).

- Points at schools
- Pointed at by documents
- Unique on school_id,code

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
code	varchar(40)		unique
name	varchar(120)		
applies_to	varchar(11)		
is_mandatory	boolean		default false
required_if_category	varchar(40)	nullable	
has_expiry	boolean		default false
is_confidential	boolean		default false
sort_order	integer		default 100
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

scheduled_jobs

7 rows

A recurring job. Deliberately not cron syntax — a school needs "every day at 02:00", not "`*/7 3-5 * * 2`", and an interval plus an hour is far easier for a non-technical administrator to read in a settings screen.

- Unique on kind

COLUMN	TYPE	NULL	NOTES
kind	varchar(48)		unique
payload	json	nullable	
enabled	boolean		default true
every_minutes	integer		default 1440
at_hour	integer	nullable	
last_run_at	timestamptz	nullable	
next_run_at	timestamptz	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Part two: tables not yet used

These are empty in the demo school. Each belongs to a feature that is built and has working endpoints - admission has taken no applications, payroll has run no cycle, nobody has requested leave. The schema is real; only the rows are missing.

Academics

substitutions empty

One period, one day, covered by somebody else (§5.7.9).

Points at employees, schools, timetable_slots

Unique on timetable_slot_id,date

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
timetable_slot_id	bigint		→ timetable_slots.id, unique
date	date		unique
absent_teacher_id	bigint		→ employees.id
substitute_teacher_id	bigint	nullable	→ employees.id
reason	text		
status	varchar(9)		
created_at	timestampz		default now()
updated_at	timestampz		default now()

Attendance

student_leave_requests empty

A guardian asking for a child to be away (§5.8.5).

Points at enrolments, schools, users

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id
from_date	date		
to_date	date		
type	varchar(9)		
reason	text		
status	varchar(9)		
requested_by	bigint	nullable	→ users.id
decided_by	bigint	nullable	→ users.id
decided_at	timestamptz	nullable	
decision_note	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Examinations and results

report_card_publications

empty

A published report card: frozen, numbered, and citing its own rules.

Points at assessment_schemes, enrolments, grading_scales, schools, users

Unique on enrolment_id,term

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id, unique
term	varchar(20)		unique
scheme_id	bigint		→ assessment_schemes.id
grading_scale_id	bigint		→ grading_scales.id
document_no	varchar(32)		
published_at	timestamptz		
published_by	bigint	nullable	→ users.id
result_status	varchar(20)		
payload	json		
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Fees and money

student_fee_plans empty

An individual override of the class default. Keyed to the enrolment, not the student: what a child is charged is a fact about a year (§3.2).

Points at enrolments, fee_plans, schools, users

Unique on enrolment_id

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
enrolment_id	bigint		→ enrolments.id, unique
fee_plan_id	bigint		→ fee_plans.id
assigned_by	bigint	nullable	→ users.id
created_at	timestampz		default now()
updated_at	timestampz		default now()

fee_periods empty

One billing month's books, open or closed.

Points at schools, users

Unique on school_id,period_year,period_month

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
period_year	integer		unique
period_month	integer		unique
status	varchar(6)		
closed_by	bigint	nullable	→ users.id
closed_at	timestampz	nullable	
note	text	nullable	
created_at	timestampz		default now()
updated_at	timestampz		default now()

HR and payroll

staff_attendance

empty

One day, one member of staff, one mark.

Points at employees, schools, users

Unique on employee_id,date

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
employee_id	bigint		→ employees.id, unique
date	date		unique
status	varchar(8)		
check_in	time without time zone	nullable	
check_out	time without time zone	nullable	
marked_by	bigint	nullable	→ users.id
corrected_by	bigint	nullable	→ users.id
remarks	varchar(200)	nullable	
created_at	timestampz		default now()
updated_at	timestampz		default now()

staff_leave_requests empty

One application, from draft to decision.

Points at academic_years, employees, leave_types, schools, users

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
employee_id	bigint		→ employees.id
leave_type_id	bigint		→ leave_types.id
academic_year_id	bigint		→ academic_years.id
from_date	date		
to_date	date		
is_half_day	boolean		default false
days	numeric(5,1)		
reason	text		
status	varchar(9)		default 'applied'
balance_exception	boolean		default false
decided_by	bigint	nullable	→ users.id
decided_at	timestamptz	nullable	
decision_note	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

leave_balances empty

What one employee has left of one type, this year.

Points at academic_years, employees, leave_types, schools

Unique on employee_id,leave_type_id,academic_year_id

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
employee_id	bigint		→ employees.id, unique
leave_type_id	bigint		→ leave_types.id, unique
academic_year_id	bigint		→ academic_years.id, unique
entitled	numeric(5,1)		
used	numeric(5,1)		default '0'
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

salary_structure_items

empty

A per-employee override of one component's value.

Points at salary_components, salary_structures, schools

Unique on structure_id,component_id

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
structure_id	bigint		→ salary_structures.id, unique
component_id	bigint		→ salary_components.id, unique
value	numeric(12,2)		
included	boolean		default true
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

payroll_runs

empty

One month's payroll, once.

Points at schools, users

Pointed at by payslips

Unique on school_id,year,month,run_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
year	integer		unique
month	integer		unique
run_no	integer		unique, default 1
is_supplementary	boolean		default false
status	varchar(10)		default 'draft'
working_days	integer	nullable	
calculated_at	timestamptz	nullable	
approved_at	timestamptz	nullable	
approved_by	bigint	nullable	→ users.id
paid_at	timestamptz	nullable	
note	text	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

payslips

empty

One person, one run. The totals are stored, not derived on read.

Points at employees, payroll_runs, schools

Pointed at by payslip_lines

Unique on run_id,employee_id · school_id,payslip_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
run_id	bigint		→ payroll_runs.id, unique
employee_id	bigint		→ employees.id, unique
payslip_no	varchar(32)		unique
working_days	integer		
lop_days	numeric(5,1)		default '0'
monthly_gross	numeric(12,2)		
total_earnings	numeric(12,2)		
total_deductions	numeric(12,2)		
net_pay	numeric(12,2)		
employer_cost	numeric(12,2)		
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

payslip_lines

empty

One component on one payslip, as it was computed.

Points at payslips, salary_components, schools

Unique on payslip_id,code

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
payslip_id	bigint		→ payslips.id, unique
component_id	bigint	nullable	→ salary_components.id
code	varchar(16)		unique
name	varchar(60)		
type	varchar(21)		
amount	numeric(12,2)		
sequence	integer		default 0
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Admission

enquiry_interactions empty

One logged contact. Append-only in practice: the follow-up history is the evidence behind a conversion number.

Points at enquiries, schools, users

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
enquiry_id	bigint		→ enquiries.id
occurred_at	timestamptz		
channel	varchar(8)		
notes	text	nullable	
outcome	varchar(23)	nullable	
by_user_id	bigint	nullable	→ users.id
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

applications

empty

One child's candidacy in one cycle.

Points at admission_cycles, applications, schools, students, users

Pointed at by admission_decisions, admission_offers, application_guardians, application_medical, application_payments, application_siblings, assessments, enquiries, interviews, message_recipients, waitlist_entries

Unique on school_id,application_no

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id, unique
cycle_id	bigint		→ admission_cycles.id
application_no	varchar(24)	nullable	unique
first_name	varchar(60)		
middle_name	varchar(60)	nullable	
last_name	varchar(60)		
date_of_birth	date		
gender	varchar(12)		
nationality	varchar(40)	nullable	
religion	varchar(40)	nullable	
caste_category	varchar(12)	nullable	
mother_tongue	varchar(40)	nullable	
place_of_birth	varchar(80)	nullable	
identification_marks	text	nullable	
is_single_child	boolean		default false
aadhaar_last4	varchar(4)	nullable	
aadhaar_verified	boolean		default false
class_applying_for	varchar(8)		
stream	varchar(20)	nullable	
second_language	varchar(40)	nullable	
optional_subject	varchar(40)	nullable	
preferred_section	varchar(4)	nullable	
admission_category	varchar(12)		default 'general'
transport_required	boolean		default false
status	varchar(27)		default 'draft'
submitted_at	timestamptz	nullable	
source	varchar(10)		default 'walk_in'
created_by	bigint	nullable	→ users.id
address	json	nullable	
previous_school	json	nullable	
declarations	json	nullable	
sibling_verified	boolean		default false

COLUMN	TYPE	NULL	NOTES
staff_ward_verified	boolean		default false
age_override_reason	text	nullable	
previous_application_id	bigint	nullable	→ applications.id
student_id	bigint	nullable	→ students.id
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

application_guardians empty

Repeatable: father, mother, guardian (§5.1.4 step 3).

Points at applications, employees, schools

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
relation	varchar(14)		
full_name	varchar(120)		
date_of_birth	date	nullable	
qualification	varchar(120)	nullable	
occupation	varchar(80)	nullable	
designation	varchar(80)	nullable	
organisation	varchar(120)	nullable	
annual_income_band	varchar(40)	nullable	
office_address	text	nullable	
mobile	varchar(20)		
alternate_mobile	varchar(20)	nullable	
email	varchar(160)	nullable	
is_primary	boolean		default false
is_emergency_contact	boolean		default false
is_authorised_for_pickup	boolean		default false
is_school_alumnus	boolean		default false
is_school_staff	boolean		default false
employee_id	bigint	nullable	→ employees.id
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()

application_siblings

empty

Step 4. A sibling already in the school is a link to a real student, never free text, because it drives a fee concession.

Points at applications, schools, students

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
student_id	bigint	nullable	→ students.id
name	varchar(120)	nullable	
age	integer	nullable	
school_name	varchar(160)	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

application_medical

empty

Step 7, in its own table because §15 requires it to be permission-gated separately from everything else on the application.

Points at applications, schools

Unique on application_id

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id, unique
blood_group	varchar(8)	nullable	
known_allergies	text	nullable	
chronic_conditions	text	nullable	
regular_medication	text	nullable	
physical_disability	text	nullable	
learning_needs	text	nullable	
vision_hearing_notes	text	nullable	
emergency_doctor	varchar(160)	nullable	
emergency_doctor_phone	varchar(20)	nullable	
consent_for_emergency_treatment	boolean		default false
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

application_payments

empty

The application fee paid against an application. Separate from `fee\_payments` because an applicant is not a student and has no enrolment.

Points at applications, schools, users

Unique on idempotency_key · school_id,receipt_no

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id, unique
application_id	bigint		→ applications.id
purpose	varchar(15)		
amount	numeric(10,2)		
method	varchar(20)		
reference	varchar(80)	nullable	
receipt_no	varchar(24)		unique
paid_at	timestampz		
collected_by	bigint	nullable	→ users.id
status	varchar(8)		default 'paid'
void_reason	text	nullable	
idempotency_key	varchar(120)	nullable	unique
created_at	timestampz		default now()
updated_at	timestampz		default now()

assessments

empty

One test or observation for one applicant.

Points at applications, schools, users

Pointed at by assessment_subjects

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
assessment_type	varchar(22)		
scheduled_at	timestamptz		
venue	varchar(80)	nullable	
seat_no	varchar(16)	nullable	
status	varchar(9)		default 'scheduled'
total_marks	numeric(6,2)	nullable	
obtained_marks	numeric(6,2)	nullable	
is_absent	boolean		default false
evaluated_by	bigint	nullable	→ users.id
remarks	text	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

assessment_subjects

empty

Subject-wise marks, for the assessments that have subjects at all.

Points at assessments, schools

Unique on assessment_id,subject

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
assessment_id	bigint		→ assessments.id, unique
subject	varchar(60)		unique
max_marks	numeric(6,2)		
obtained	numeric(6,2)	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

interviews empty

An admission interview: when, with whom, and what was concluded.

Points at applications, schools, users

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
scheduled_at	timestamptz		
venue	varchar(80)	nullable	
panel_member_ids	json	nullable	
status	varchar(9)		default 'scheduled'
structured_scores	json	nullable	
child_rating	integer	nullable	
parent_rating	integer	nullable	
recommendation	varchar(12)	nullable	
notes	text	nullable	
conducted_by	bigint	nullable	→ users.id
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

admission_decisions empty

Append-only. A reversal is a new row, not an edit of the old one — the history of who decided what, and why, is the whole point (§5.1.9(13)).

Points at applications, schools, users

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
decision	varchar(10)		
decided_by	bigint	nullable	→ users.id
decided_at	timestamptz		
reason	text		
seat_category	varchar(20)	nullable	
conditions	text	nullable	
over_allocation_approved	boolean		default false
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

admission_offers empty

An offer of a place made to an applicant, with its expiry and whether it was taken up.

Points at applications, schools

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
application_id	bigint		→ applications.id
offered_at	timestamptz		
expires_on	date		
offer_amount	numeric(10,2)	nullable	
status	varchar(9)		default 'issued'
accepted_at	timestamptz	nullable	
released_at	timestamptz	nullable	
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

waitlist_entries empty

An ordered queue per class, not a label on an application.

Points at admission_cycles, applications, schools

Unique on application_id · cycle_id,class_name,rank

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
cycle_id	bigint		→ admission_cycles.id, unique
class_name	varchar(8)	unique	
application_id	bigint		→ applications.id, unique
rank	integer	unique	
status	varchar(9)		default 'waiting'
id	bigint		primary key
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

Communication

messages

empty

One send: a body, an audience, and a status.

Points at message_templates, schools, users

Pointed at by message_recipients, notices

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
category	varchar(11)		
channel	varchar(8)		default 'email'
template_id	bigint	nullable	→ message_templates.id
template_version	integer	nullable	
subject	varchar(200)		
body	text		
audience	json		
status	varchar(9)		default 'draft'
scheduled_for	timestamptz	nullable	
sent_at	timestamptz	nullable	
created_by	bigint	nullable	→ users.id
approved_by	bigint	nullable	→ users.id
approved_at	timestamptz	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

message_recipients

empty

One person, one address, one delivery outcome.

Points at applications, guardians, messages, schools, students, users

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
message_id	bigint		→ messages.id
user_id	bigint	nullable	→ users.id
guardian_id	bigint	nullable	→ guardians.id
student_id	bigint	nullable	→ students.id
application_id	bigint	nullable	→ applications.id
to_address	varchar(200)		
channel	varchar(8)		default 'email'
context	json		
status	varchar(9)		default 'queued'
attempts	integer		default 0
sent_at	timestamptz	nullable	
failure_reason	text	nullable	
provider_ref	varchar(120)	nullable	
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

notification_preferences

empty

One opt-out, per person per category per channel.

Points at schools, users

Unique on user_id,category,channel

COLUMN	TYPE	NULL	NOTES
id	bigint		primary key
school_id	bigint		→ schools.id
user_id	bigint		→ users.id, unique
category	varchar(11)		unique
channel	varchar(8)		unique, default 'email'
opted_out	boolean		default true
created_at	timestamptz		default now()
updated_at	timestamptz		default now()

System

jobs

empty

The background work queue - messages to send, report cards to build. Drained by `worker.py`.

Points at schools
Unique on idempotency_key

COLUMN	TYPE	NULL	NOTES
school_id	bigint		→ schools.id
kind	varchar(48)		
payload	json	nullable	
status	varchar(9)		default 'pending'
run_after	timestampz		
started_at	timestampz	nullable	
finished_at	timestampz	nullable	
attempts	integer		default 0
max_attempts	integer		default 3
last_error	text	nullable	
result	json	nullable	
idempotency_key	varchar(120)	nullable	unique
requested_by	bigint	nullable	
id	bigint		primary key
created_at	timestampz		default now()
updated_at	timestampz		default now()